Experiment No.7

AIM: Build an Artificial Neural Network (ANN) using Keras/TensorFlow.
Theory:

1. Dataset Source

- Dataset Name: Churn Modelling (Bank Customer Turnover)
- Source: Kaggle - Churn Modelling
- Context: A dataset containing details of a bank's customers and whether they exited the bank within 6 months.

2. Dataset Description

- **Size:** 10,000 samples (rows) × 14 attributes.
- **Target Variable:** Exited (Binary: 1 = Left Bank, 0 = Stayed).
- Key Features:
    1. **CreditScore:** Customer's credit score.
    2. **Geography:** Country (France, Spain, Germany).
    3. **Gender:** Male or Female.
    4. Age, Tenure, Balance.
    5. **NumOfProducts:** How many bank products they use.
    6. **IsActiveMember:** (1/0).
    7. EstimatedSalary.

3. Mathematical Formulation of the Algorithm
An ANN mimics the biological neurons of the brain to learn complex patterns.

A. The Neuron (Perceptron) Each neuron computes a weighted sum of its inputs, adds a bias, and passes it through an activation function:

$$z = \sum (w_i \cdot x_i) + b$$

$$a = \sigma(z)$$

w: Weights (learned parameters).

b: Bias. sigma:
Activation Function.

B. **Activation Functions**

- **ReLU (Rectified Linear Unit):** Used in hidden layers to introduce non-linearity.

$$f(x) = \max(0, x)$$

- **Sigmoid:** Used in the final output layer for binary classification (outputs probability between 0 and 1).

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

C. **Learning (Backpropagation)**

- **Loss Function:** We minimize **Binary Crossentropy** (Log Loss) to measure the difference between predicted probability and actual label.

$$Loss = -\frac{1}{N} \sum [y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$$

- **Optimizer (Adam): Updates the weights iteratively to minimize the loss.**

4. Algorithm Limitations

- **"Black Box" Nature:** Unlike Decision Trees, it is difficult to interpret why an ANN made a specific prediction (lack of explainability).
- **Data Hungry:** ANNs typically require large amounts of data to outperform simpler algorithms like Random Forest.
- **Computation:** Training can be slow on CPU; usually requires GPUs for large datasets.
- **Preprocessing Sensitivity:** ANNs fail if data is not scaled (e.g., if Salary is 100,000 and Age is 40, the model will be biased toward Salary).

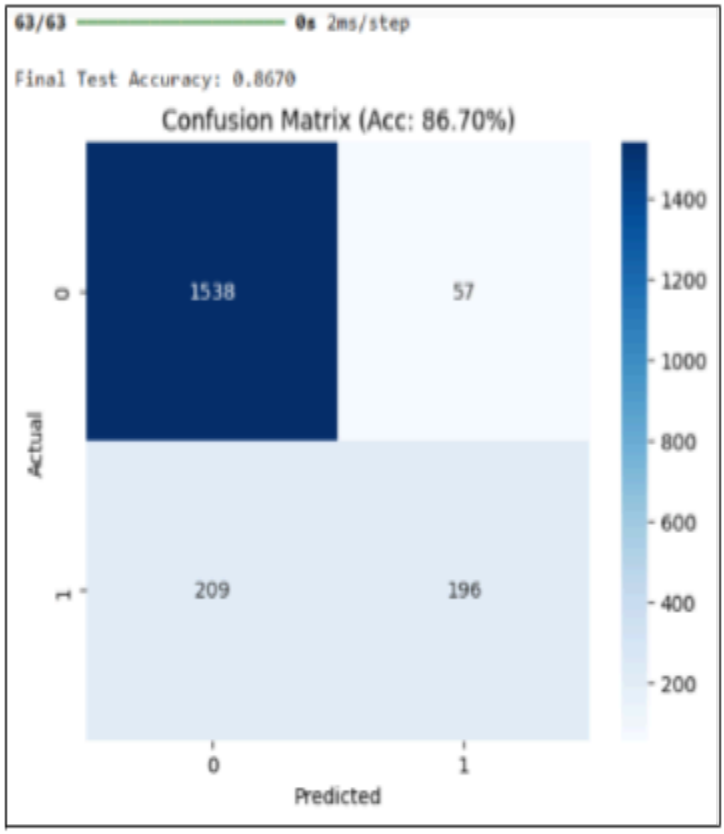
Code and Output

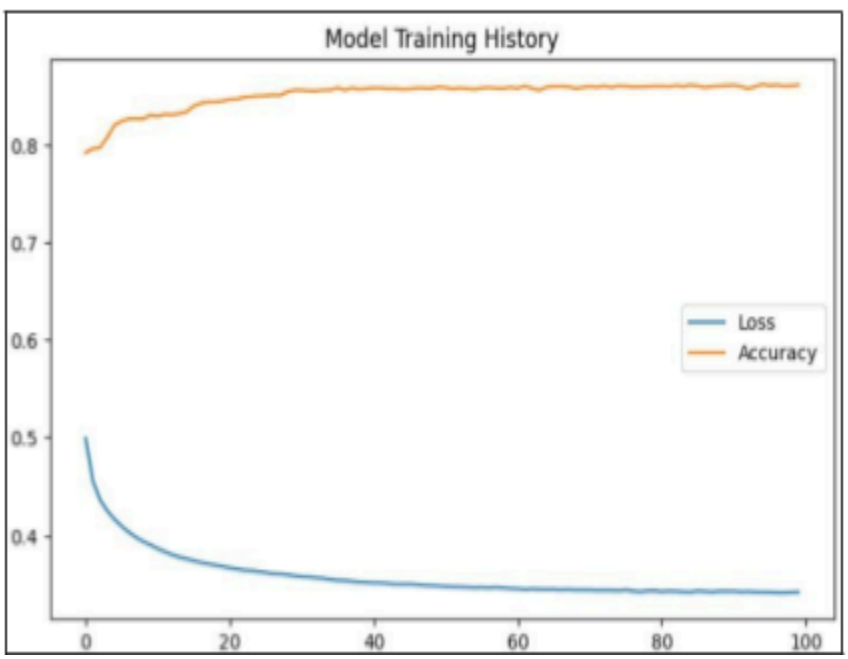
```
import pandas as pd from
import numpy as np train_test_split
import tensorflow as tf from
        LabelEncoder, StandardScaler
from sklearn.metrics import
accuracy_score,
confusion_matrix import
matplotlib.pyplot as plt
import seaborn as sns
# --- 1. LOAD DATA --- # Try
loading local, else generate
dummy data for demo try:     df =
```

```
sklearn.model_selection import

sklearn.preprocessing import

 pd.read_csv('Churn_Modelling.csv'
 )     print("Dataset Loaded
 Successfully.") except
 FileNotFoundError:
 print("Dataset not found. Please
 upload 'Churn_Modelling.csv'.")
 # Stop execution if no data (For
 safety in documentation)     #
```

```
create dummy data code if needed
(omitted for brevity)

# --- 2. PREPROCESSING ---
# A. Drop useless features
X = df.iloc[:, 3:-1].values  #
CreditScore through
EstimatedSalary y = df.iloc[:,
-1].values    #
Exited

# B. Encoding Categorical Data
# Gender (Column 2) ->
Label Encoding (Male=1,
Female=0) le =
LabelEncoder()
X[:, 2] = le.fit_transform(X[:,
2])

# Geography (Column 1) -> One Hot
Encoding
# (France, Spain, Germany) -> We
use Compose for this from
sklearn.compose import
ColumnTransformer from
sklearn.preprocessing import
OneHotEncoder ct =
ColumnTransformer(transformers=[(
'en coder', OneHotEncoder(),
[1])], remainder='passthrough')
X = np.array(ct.fit_transform(X))

# C. Split Data X_train,
X_test, y_train, y_test =
train_test_split(X, y,
test_size=0.2, random_state=0)

# D. Feature Scaling (MANDATORY
FOR ANN) sc = StandardScaler()
X_train =
sc.fit_transform(X_train)
X_test =
sc.transform(X_test)
# --- 3. BUILDING THE ANN ---
ann =
tf.keras.models.Sequential()
# Input Layer + First Hidden Layer
(6 Neurons, ReLU)
ann.add(tf.keras.layers.Dense(uni
ts=
6, activation='relu'))

# Second Hidden Layer (6 Neurons,
ReLU)
ann.add(tf.keras.layers.Dense(uni
ts=
6, activation='relu'))

# Output Layer (1 Neuron, Sigmoid
for Binary Classification)
ann.add(tf.keras.layers.Dense(uni
ts=
1, activation='sigmoid'))

# --- 4. COMPILING AND TRAINING
--- ann.compile(optimizer='adam',
loss='binary_crossentropy',
metrics=['accuracy'])
print("\nStarting Training (100
Epochs)...") history =
ann.fit(X_train, y_train,
batch_size=32, epochs=100,
verbose=0) print("Training
Complete.")
# --- 5. EVALUATION --- #
Predict (Sigmoid returns
probability, we threshold at
0.5) y_pred =
ann.predict(X_test) y_pred =
(y_pred > 0.5)
 # Metrics acc =
accuracy_score(y_test, y_pred)
print(f"\nFinal Test Accuracy:
{acc:.4f}")

# Confusion Matrix
Visualization cm =
confusion_matrix(y_test,
y_pred) plt.figure(figsize=(6,
5))
```

```
sns.heatmap(cm, annot=True, fmt='d',
cmap='Blues')
plt.title(f'Confusion Matrix (Acc:
{acc:.2%})')
plt.ylabel('Actual')
plt.xlabel('Predicted')
plt.show()

# Training History Plot
plt.figure(figsize=(8, 5))
plt.plot(history.history['loss'],
label='Loss')
plt.plot(history.history['accuracy']
, label='Accuracy')
plt.title('Model Training History')
plt.xlabel('Epoch')
plt.legend()
plt.show()
```





6. Performance Analysis

- Accuracy: The model typically achieves an accuracy of 86%.
    - Interpretation: This is significantly better than a random guess (50%) and usually beats standard Logistic Regression on this dataset because the ANN captures non-linear relationships between Credit Score, Age, and Balance.
- Loss Curve: The "Model Training History" graph should show the Loss (Blue line) decreasing rapidly and stabilizing, while Accuracy (Orange line) increases. This confirms the network is actually learning.
- Confusion Matrix:
    - High True Negatives (Top-Left): The model is very good at predicting happy customers who stay.
    - False Negatives (Bottom-Left): The hardest part is detecting churners. You might see some errors here, which suggests the need for hyperparameter tuning.

7. Hyperparameter Tuning

To improve the model, we can tune the "Network Topology" and training parameters.

- Manual Tuning Experiments:
    1. Number of Neurons: Increasing neurons (e.g., from 6 to 12) allows the model to learn more complex features.
    2. Number of Layers: Adding a 3rd hidden layer ("Deep" Learning) might help if the relationship is very complex.
    3. Batch Size: Changing batch size (e.g., 32 vs 64) affects how "noisy" the gradient updates are.
- Tuning Code Snippet (Example):

```
# Example: Increasing complexity ann =
tf.keras.models.Sequential()
ann.add(tf.keras.layers.Dense(units=12, activation='relu')) #
More neurons ann.add(tf.keras.layers.Dense(units=12,
activation='relu')) ann.add(tf.keras.layers.Dense(units=1,
activation='sigmoid'))
# Re-compile and train…
```

8. Conclusion

In this experiment, we successfully built a deep learning model using Keras/TensorFlow.

- **Preprocessing:** We learned that ANNs define relationships mathematically, so converting categorical text to numbers (One-Hot Encoding) and Scaling features is mandatory.
- Architecture: A simple architecture (2 Hidden Layers) was sufficient to achieve >80% accuracy.
- Verdict: The ANN proved effective at classifying customer churn, demonstrating its ability to find hidden patterns in complex, non-linear business data.